

Josef Krivan
s315310
02/02/2021

Project Proposal

General Idea

The project I decided on was a rather simple one. I would like to monitor the temperature inside my apartment. This will involve using a sensor connected to an IOT device to record such data. I intend to use a phone (maybe a PC) to monitor this data in real time.

The Purpose

The reason I chose this project is for 2 reasons.

One, monitoring temperature data in my home is something I will certainly end up continuing to do after the project is complete as it will be useful all year round.

Two, it appears to be a relatively simple project. Since I have zero experience programming for hardware outside of my own Windows environment, I already see this as a challenge to learn something new (learning how to create a mobile app/program for an IOT device).

Technical Details

Hardware

For the recording of data, I plan on using a Raspberry PI, mainly because it appears easier to learn, and it will be easier for me to work with since I already know the basics of Python. Another nice thing is the fact that it is basically a mini-PC, with support for Linux distros. It also supports Windows 10 IOT core, which sounds great, but I question the support of it and if there is a sizeable community developing on it.

I did not consider an Arduino at first, mainly because that is more of a microcontroller, and I am only looking to monitor data, not modify it. It is also programmed using C++, which I have no knowledge for. If someone does recommend it for this type of project, then I will consider it.

For the monitoring of data, I will be using an Android device, that should connect to the internet and fetch the data from the IOT device. I'm not certain on how this will exactly work yet, and might require some assistance.

Software

I plan on creating a simple Android application that can fetch the data from the IOT device via Wi-Fi. The app's main feature will contain a visual time series to see recent temperature history. I am not sure on the technical details of this app just yet and will probably need help on it.

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Time Plan/Allocation

Making a time plan is more difficult considering the current circumstances. First will I need to confirm which IOT device I will use, and since money is tight, I hope purchasing such a device will not be too expensive. Receiving it as early as possible, so I can start working with it.

I will not follow a strict time plan, but rather a set of goals that should be completed in order.

First I need to setup the IOT device, along with the sensor, making sure it works on the device. Then I need to figure out how to capture the data and send it via Wi-Fi.

Afterwards, I will need to allocate a significant amount of time to not only learn how to develop apps for Android, but to also create the simple app as planned.

The last step is to connect these 2 devices via the network and have the ability to observe the data captured, and obtain some results.

I plan on leaving a good amount of time for the report, starting it as early as possible and keeping up with current project progress.

What to Expect

By the end of this project, I hope to have a prototype of my IOT project. I hope to be able to capture data and display it on a graph via a phone. If this is successful, I may add some extra features to the app (notification if its too hot or cold etc...)

With the limited skills I have, I hope to be able to complete this project at a functional level.